

# Assignment 1: E-Purchase Business Process Modeling

BPMN 2.0 and Petri-net models for an Amazon/Souq-like order process

## Submission focus

This report models the same business process in two accepted ways: BPMN 2.0 for business readability and Petri nets for formal token-based behavior. It includes the as-is process, a to-be improvement, issue analysis, and exported evidence files.

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## 1 Executive Scope

The modeled process covers an e-purchase from order submission through validation, stock handling, payment, fulfillment, delivery, complaint handling, and final archiving. To maximize assignment coverage, the same process is documented in both required modeling styles:

- BPMN 2.0 models for the as-is and to-be business process.
- Petri-net models for the same as-is and to-be behavior, including choices, synchronization, loops, and final markings.

The to-be design improves early validation, parallelizes independent checks, connects complaints to the same order case, and adds proactive SLA monitoring for delayed goods.

## 2 Prepared Evidence Assets

| File                                                   | Purpose                                                                  |
|--------------------------------------------------------|--------------------------------------------------------------------------|
| output/assignment1_epurchase_as_is.bpmn                | BPMN 2.0 source for the as-is model, importable into bpmn.io or Camunda. |
| output/assignment1_epurchase_to_be.bpmn                | BPMN 2.0 source for the improved to-be model.                            |
| output/assignment1_epurchase_as_is_petri.dot and .pnml | Graphviz and PNML versions of the formal as-is Petri net.                |
| output/assignment1_epurchase_to_be_petri.dot and .pnml | Graphviz and PNML versions of the improved to-be Petri net.              |
| screenshots/assignment1/*.svg                          | Browser/tool exports used as the visual evidence in this report.         |

## 3 Evaluation Criteria Coverage

The assignment rubric is used as an explicit checklist. The report covers each marking category as follows:

| Criteria category     | Marks | How this report covers it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Process model quality | 20    | BPMN 2.0 models use start/end events, tasks, exclusive gateways, parallel split/join gateways, event-based gateways, timer/message intermediate events, swimlanes, and logical sequence flows. Petri nets also model the process with places, transitions, arcs, initial marking, final marking, choices, joins, and loops. Core e-purchase steps are shown: order capture, stock check, address validation, payment authorization, invoice, payment receipt, packing, shipping, delivery, complaint, and archive. |
| Tool proficiency      | 5     | BPMN source files include lane metadata and are prepared for bpmn.io/Camunda import. Petri-net DOT/PNML files are prepared for Graphviz, diagrams.net, or PNML tools. The submitted report includes SVG exports rendered into the PDF through the verified cache.                                                                                                                                                                                                                                                  |

|                               |    |                                                                                                                                                                                                                                                                                         |
|-------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use of intermediate events    | 10 | The as-is BPMN uses an intermediate delivery event and a timer event for delayed goods. The to-be BPMN uses an SLA event and SLA-breach branch. Both models clearly route delayed goods into complaint registration, customer notification/escalation, and return to delivery tracking. |
| Documentation and clarity     | 15 | Every diagram is labeled, introduced, and followed by text explaining what the model means. Participants, responsibilities, formal Petri-net summary, file evidence, and case coverage are documented.                                                                                  |
| Analysis before enhancement   | 10 | The as-is issue register identifies reactive complaints, late address correction, weak synchronization, separated complaint ownership, and late archiving. Each issue includes consequence, type, and justification through its proposed fix.                                           |
| Enhancement suggestions       | 10 | The to-be model moves validation earlier, parallelizes stock/payment checks, standardizes fulfillment, adds tracking, connects complaints to the order case, and makes delay handling proactive.                                                                                        |
| Comparative analysis          | 10 | The before/after comparison explains what changed and why. Expected outcomes are added for shorter waits, lower rework, clearer customer communication, and stronger auditability.                                                                                                      |
| Report quality and submission | 10 | The report has a cover page, contents, list of figures, consistent table styling, labeled diagrams, and no LaTeX-drawn diagrams. The repository is cleaned after compilation.                                                                                                           |
| Originality and effort        | 10 | The process is a realistic Amazon/Souq-like scenario with payment service, warehouse, courier, support, complaint escalation, SLA tracking, audit events, and two modeling methods rather than only the minimum model.                                                                  |

## 4 Modeling Standard and Core Step Checklist

| Requirement                     | Implementation evidence                                                                                                                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Correct BPMN start/end behavior | The as-is model starts at <i>Order received</i> and reaches either <i>Order rejected</i> or <i>Order closed</i> . The to-be model starts at <i>Checkout submitted</i> and has the same accepted/rejected closure logic.                                         |
| Core e-purchase flow            | Both versions include order validation, stock decision, payment authorization, invoice emission, payment receipt, warehouse fulfillment, shipment, delivery confirmation, and archiving.                                                                        |
| Decision logic                  | Exclusive gateways model stock availability, address/order validity, merge points, and rejection paths. Parallel gateways model only true independent split/join behavior, so mutually exclusive address paths are merged before the payment/fulfillment split. |
| Goods-delay event               | The delayed-goods path is not only mentioned in text; it is modeled as an event-driven complaint/escalation branch.                                                                                                                                             |
| Petri-net formal semantics      | The Petri nets include initial/final markings, routing transitions, parallel split/join behavior, and a loop from delay escalation back to delivery waiting/tracking.                                                                                           |
| Online-tool evidence            | SVG exports are stored in <code>screenshots/assignment1/</code> , while importable BPMN, DOT, and PNML source files remain in <code>output/</code> .                                                                                                            |

## 5 Lecture Concepts Applied

The report follows the lecture definitions of a business process as structured work involving people, systems, data, and resources to deliver value to a customer. The BPMN lecture is used for events, tasks, gateways, sequence flows, and process readability. The Petri-net lecture is used for places, transitions, arcs, markings, reachability, synchronization, and loops. The process-analysis lecture is used for issue registers, customer interfaces, quality risks, cycle-time reduction, and improvement opportunities.

## 6 BPMN 2.0 Model

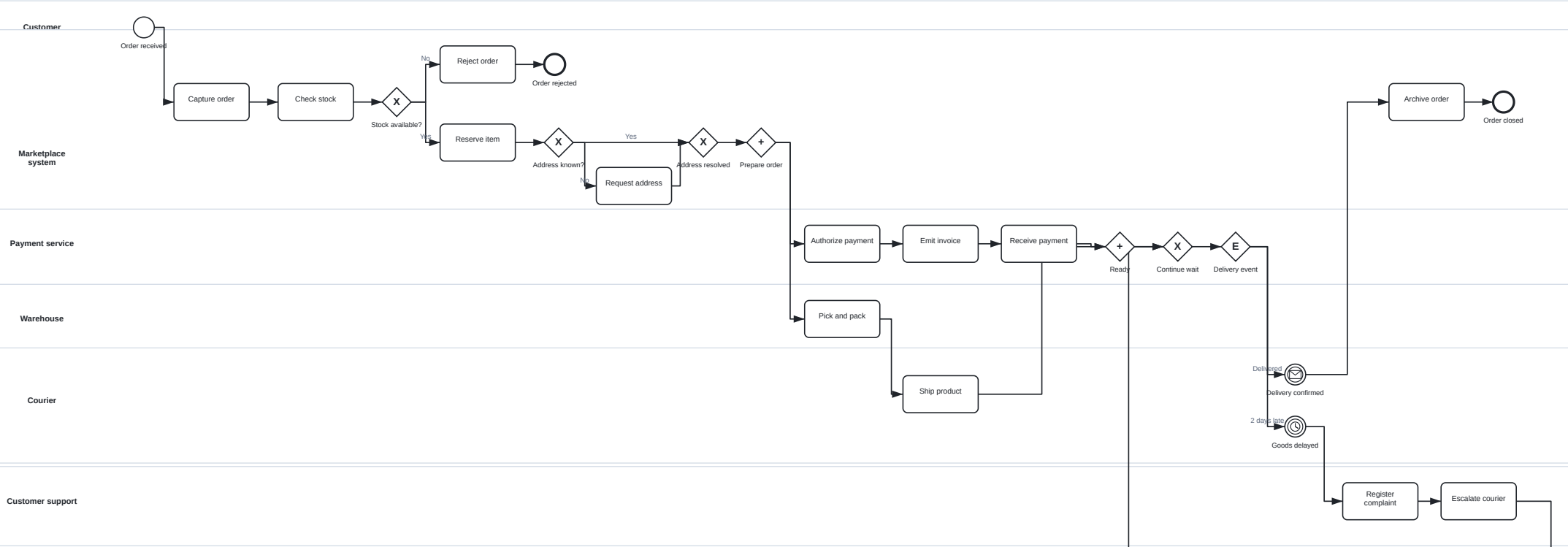
### 6.1 Participants and Responsibilities

| Participant or lane | Responsibility                                                                                                   |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Customer            | Places the order, provides address data, receives invoice and delivery, and may complain when goods are delayed. |
| Marketplace system  | Captures the order, checks stock, validates data, emits invoice, tracks status, and archives the order.          |
| Payment service     | Authorizes payment and returns payment status to the marketplace process.                                        |
| Warehouse           | Reserves stock, picks, packs, and prepares the physical shipment.                                                |
| Courier             | Ships the product and reports delivery or delay events.                                                          |
| Customer support    | Registers complaints, notifies customers, and escalates delayed courier cases.                                   |

### 6.2 As-Is BPMN

The as-is process starts when the order is received. The system captures the order and checks stock. If stock is unavailable, the order is rejected. If stock is available, the item is reserved, address data is checked, and missing address data is requested. The two address outcomes merge through an exclusive gateway before the process splits into parallel payment and warehouse work. The payment branch authorizes payment, emits the invoice, and receives payment; the fulfillment branch picks, packs, and ships the product. After both branches synchronize, the process waits for a delivery event.

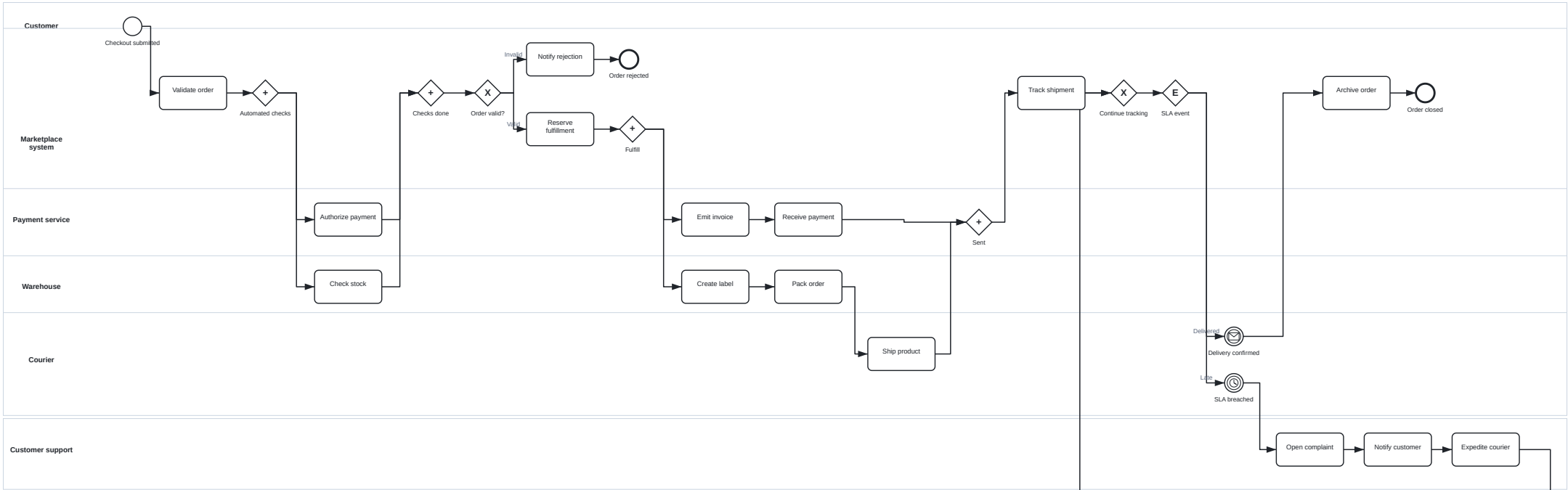
The delayed-goods scenario is represented by an intermediate timer event after shipment. If delivery confirmation is not received within the SLA, the timer branch registers a complaint, escalates the courier, and loops back to a merge before the event-based delivery gateway. The delivery confirmation event therefore remains a single catch event owned by the delivery wait.



### 6.3 To-Be BPMN

The to-be process moves validation earlier, runs stock and payment checks in parallel, standardizes fulfillment preparation, adds explicit payment receipt, adds shipment tracking, and makes the complaint path proactive. The SLA breach event opens a complaint automatically, notifies the customer, expedites the courier, and then loops back through a tracking merge before the next SLA/delivery wait.





## 7 Petri-Net Model

The Petri-net model is the second modeling method. States are represented as places, activities and routing events as transitions, and allowed movement as directed arcs. One token starts in the initial place. Silent transitions are used only for routing choices, parallel splits, and joins.

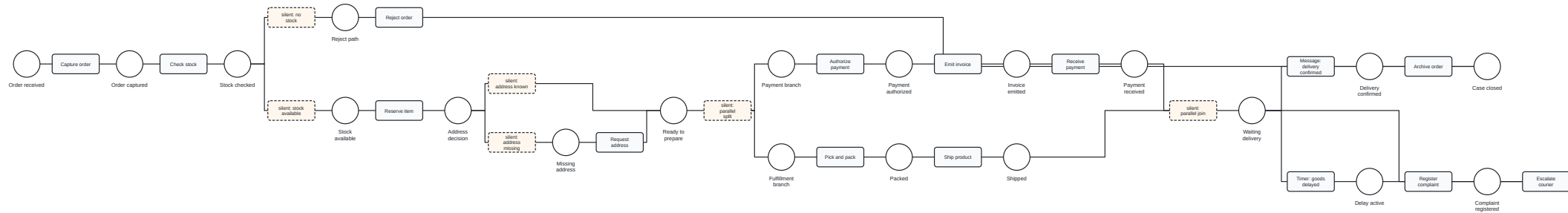
### 7.1 Formal Net Summary

| Model | Places | Transitions | Arcs | Initial and final markings                   |
|-------|--------|-------------|------|----------------------------------------------|
| As-is | 20     | 21          | 44   | $M_0 = \{p\_start\}$ and $M_f = \{p\_done\}$ |
| To-be | 24     | 23          | 50   | $M_0 = \{p\_start\}$ and $M_f = \{p\_done\}$ |

### 7.2 As-Is Petri Net

The as-is Petri net has one token in **p\_start**. The stock decision place enables either rejection or reservation. Both address outcomes produce the same preparation place, matching the corrected BPMN exclusive merge. The preparation split produces one token for the payment branch and one token for the fulfillment branch; the join requires payment receipt and shipment before delivery waiting can continue. The goods-delay timer transition creates the complaint token, and the escalation transition loops back to the delivery waiting place.

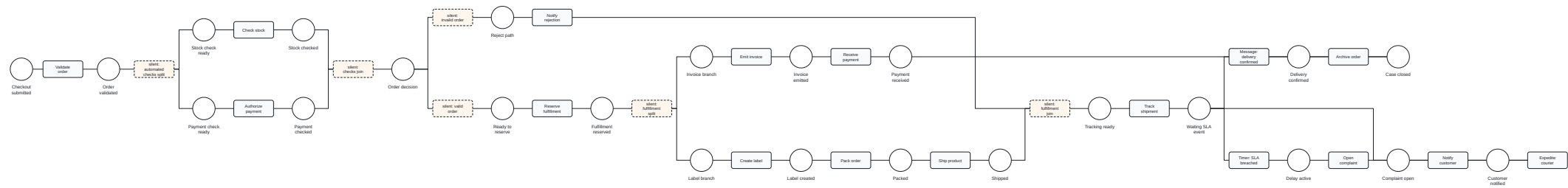
Assignment 1 As-Is Petri Net



### 7.3 To-Be Petri Net

The to-be Petri net shows two improvements formally. First, stock check and payment authorization are enabled together after validation, then synchronized before the order validity decision. Second, invoice/payment receipt and physical fulfillment are split into parallel branches and joined before tracking. The SLA breach transition sends the case through complaint opening, customer notification, and courier expedition before returning to the SLA wait place.

Assignment 1 To-Be Petri Net



## 8 As-Is Analysis

| No. | Issue                      | Consequence                                                | Type               | Proposed fix                                                  |
|-----|----------------------------|------------------------------------------------------------|--------------------|---------------------------------------------------------------|
| 1   | Complaint is reactive      | Customer may discover the delay before the company reacts. | Customer interface | Add proactive shipment tracking and SLA monitoring.           |
| 2   | Address correction is late | Reserved stock may wait while customer data is fixed.      | Data/process       | Validate address before reservation.                          |
| 3   | Weak synchronization       | Invoice or shipment may wait for manual status checking.   | IT/process         | Automate parallel checks and clear joins.                     |
| 4   | Complaint path is separate | Support may reconstruct context manually.                  | Organization       | Attach complaint to the same order case and assign ownership. |
| 5   | Archiving only at closure  | Monitoring data is late and hard to mine.                  | IT                 | Record status events throughout execution.                    |

## 9 Enhancements and Comparative Analysis

| Area             | As-is process                                     | To-be improvement                                                     |
|------------------|---------------------------------------------------|-----------------------------------------------------------------------|
| Checkout data    | Address may be requested after stock reservation. | Validate address and order data before stock is held.                 |
| Cycle time       | Several checks can wait sequentially.             | Run stock and payment checks in parallel.                             |
| Delay handling   | Complaint begins after a late-delivery event.     | SLA monitoring opens the complaint and notification path proactively. |
| Customer quality | Customer may need to ask for status.              | Tracking events and notifications are part of the process.            |
| Cost and rework  | Support may manually reconstruct order context.   | Complaint is connected to the same order case.                        |
| Modeling rigor   | BPMN shows business readability.                  | Petri nets add formal token semantics for choices, joins, and loops.  |

## 9.1 Expected Outcomes from the To-Be Model

| Improvement target                 | Change made                                                         | Expected outcome                                                                                          |
|------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Shorter waiting before fulfillment | Validate order data first and run stock/payment checks in parallel. | Fewer orders reserve inventory while still missing data; independent checks no longer wait unnecessarily. |
| Faster delay response              | Add SLA monitoring and proactive complaint opening.                 | The company reacts before the customer has to chase the order, improving customer experience.             |
| Lower manual rework                | Keep complaint handling attached to the same order case.            | Support sees order, shipment, and complaint context together instead of reconstructing the case manually. |
| Better audit trail                 | Record status events before final archiving.                        | Later process mining can identify bottlenecks, courier delays, and repeated failure points.               |

## 10 Case Coverage

- Normal order: checkout, validation, stock/payment checks, fulfillment, delivery confirmation, and archive.
- Rejected order: invalid payment, invalid data, or unavailable stock leads to rejection.
- Missing address: handled before stock is blocked in the to-be model.
- On-time delivery: delivery message event continues to archive.
- Goods delayed: timer/SLA event opens complaint, notifies customer, escalates courier, and loops back to delivery tracking.
- Audit trail: status updates and archiving support later monitoring and process mining.

## 11 Conclusion

The BPMN models provide the business-readable view required for process discussion, while the Petri nets provide the formal execution view required for reasoning about choices, concurrency, synchronization, and loops. The to-be model directly addresses the main weaknesses in the as-is process: late validation, reactive complaint handling, and weak status visibility.